

Predicting peptide-HLA complex stability with a multilayer perceptron

Technical report · Rasmussen et al. dataset · 3 October 2026

A fully connected network regresses the logarithm of complex half-life from the stacked peptide and HLA pseudosequence. Trained independently on five constrained splits, it reaches a test RMSE of 1.17 ± 0.05 log units and a Spearman correlation of 0.770 ± 0.021 , against 1.83 for a train-mean baseline. Error grows from 1.00 on test rows whose peptide and HLA both appear in training to 1.64 on rows carrying an HLA the model has never seen. The log offset ϵ was chosen by experiment (section 9): at 0.1 it improves rank correlation on all five splits relative to the 0.001 used initially.

1. Data and splits

The source is `DATA/rasmussen_et_al_dataset.xlsx`, 28,166 measured peptide-HLA complexes with columns `allele`, `peptide` (9 residues), `half_hours`, `hla_seq` (182 residues) and `hla_pseudoseq` (34 residues). It contains 5,002–8,751 distinct peptides depending on the split and only 74 distinct HLA pseudosequences.

Five train/test splits were produced beforehand by `split_dataset.py` and stored as NumPy archives. Each split is an independent 70/30 partition of all rows, constrained so that (i) at least 5 HLA sequences and 100 peptides occur only in the test set, and (ii) the train and test means of $\ln(x)$ and $\ln(x)^2$ differ by at most 0.05 and 0.10. All five splits met both moment constraints on their first seed. They are not cross-validation folds: the holdouts are drawn independently, so test sets overlap between splits.

Table 1 — Row budget, identical for every split.

Partition	Rows	Share	Role
Training	17,744	63.0%	Gradient steps
Validation	1,972	7.0%	Early stopping only
Testing	8,450	30.0%	Final scoring only
Total	28,166	100%	

The validation set was not part of the original splits; it is carved out of each training set inside the training script, by a seeded random permutation of rows (`--validation-fraction 0.1`). Its only use is to decide when to stop and which weights to keep. The test sets are read once, after training ends.

The validation draw is random over rows, so validation peptides and HLAs also occur in training. Validation loss is therefore easier than test loss and is not a generalization estimate — only the test figures in section 7 are.

2. Feature encoding

Each example is the peptide stacked with the HLA pseudosequence, one-hot encoded:

- **Peptide:** 9 positions, every peptide in the dataset is a 9-mer.
- **HLA:** the 34-residue pseudosequence, not the full 182-residue chain. The pseudosequence keeps the residues lining the binding groove, so every input position is informative. With only 74 distinct

HLA sequences in the data, the longer form would let the network treat the HLA block as an allele identifier and memorize it, which is precisely what fails on unseen alleles.

- **Alphabet:** the 20 standard amino acids (ACDEFGHIKLMNPQRSTVWY); an unrecognised residue raises an error rather than being silently encoded as zeros.

This gives $(9 + 34) \times 20 = \mathbf{860 \text{ binary input features}}$. The encoding carries no notion of residue similarity: leucine and isoleucine are as unrelated as leucine and aspartate. A BLOSUM62 encoding, which would supply that similarity at the same input width, is the natural next experiment (section 10).

3. Target transform and zero handling

Half-lives span 0 to 256.7 hours, are strongly right-skewed (median 0.3 h among the non-zero values), and 5,679 rows — 20% of the dataset — are exactly zero. Regressing the raw value under a squared-error loss would let a handful of long-lived complexes dominate the gradient.

The model therefore predicts

$$y = \ln(\text{thalf_hours} + \epsilon), \quad \epsilon = 0.1$$

Adding ϵ inside the logarithm is the regularization of the singularity at zero: it is smooth and monotone over the whole range, unlike clamping, and it keeps the 20% of zero rows at a single finite value, $\ln(0.1) \approx -2.30$. It is exposed as `--epsilon`.

The value 0.1 h is the reporting resolution of the assay — half-lives are given to one decimal — so it encodes the statement that the data cannot distinguish a zero from any value below the detection limit. Section 9 reports the experiment behind the choice: an earlier draft of this work used $\epsilon = 0.001$, which placed the zero rows 4.6 log units below a typical 0.1 h measurement and let them dominate the loss.

Targets are then standardized with the *training* mean and standard deviation of each split, which conditions the optimization; predictions are mapped back before scoring, so every error in this report is in natural-log units of hours.

ϵ sets how far the zero spike sits below the rest of the distribution, so it rescales every error in this report. Absolute errors are therefore comparable only between runs sharing the same ϵ , and not with published figures that use a different transform.

4. Architecture

A fully connected feed-forward network, ReLU activations, dropout after each hidden layer, single linear output. Three weight layers in total: two hidden, one output.

Table 2 — Layer stack and parameter count.

#	Layer	Units in	Units out	Activation	Parameters
—	Input (one-hot)	—	860	—	0
1	Linear + ReLU + Dropout 0.2	860	256	ReLU	220,416
2	Linear + ReLU + Dropout 0.2	256	128	ReLU	32,896
3	Linear (output)	128	1	identity	129
Total trainable					253,441

The widths are set by `--hidden 256 128` and were not tuned; they are a deliberate baseline. The parameter count is about 14 times smaller than the 3.55 million distinct (peptide, HLA) pairs the training rows could in principle encode, but still large relative to 17,744 training rows, which is why the regularization below matters.

5. Training and regularization strategy

Four regularizing mechanisms act together; the first is about the target, the other three about the weights.

Target ϵ	0.1 added before the logarithm (section 3), removing the singularity at zero without discarding the 20% of zero-stability rows.
Dropout	$p = 0.2$ after each hidden layer, active in training only.
Weight decay	1×10^{-5} , applied by Adam to all parameters.
Early stopping	Validation MSE is evaluated every epoch; the weights of the best epoch are restored at the end. Training halts after 25 epochs without improvement, or at 300 epochs.

Table 3 — Optimization settings, identical across splits.

Setting	Value	Setting	Value
Optimizer	Adam	Loss	MSE on standardized target
Learning rate	1×10^{-3}	Batch size	256
Weight decay	1×10^{-5}	Max epochs	300
Dropout	0.2	Patience	25 epochs
Target ϵ	0.1 h	Target scaling	Train mean and SD
Seed (split i)	$0 + i$	Device	CPU

No learning-rate schedule and no gradient clipping are used; the gradient norm is measured each step but left unmodified. Each split trains in 4-12 seconds on CPU, and runs are reproducible given the seed.

6. Convergence

Figure 1 shows, per split, the training and validation MSE in log units and the mean gradient L2 norm per epoch on a logarithmic axis. All five runs converge in the same way: the gradient norm rises over the first ~10 epochs as the weights leave their initialization, then decays monotonically; training loss flattens; validation loss reaches a plateau near 1.0–1.2 and then drifts upward as the network begins to overfit, which is where early stopping intervenes.

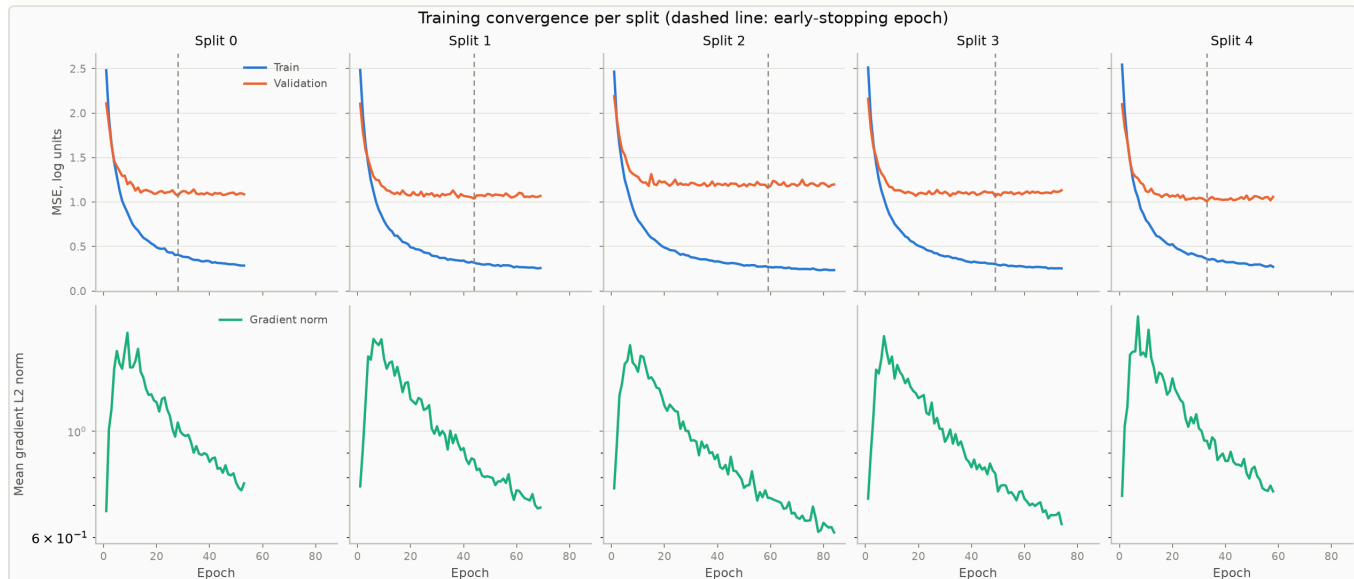


Figure 1. Convergence per split. Top: ■ training and ■ validation MSE, log units, shared scale. Bottom: ■ mean gradient L2 norm per epoch, logarithmic, shared scale. Dashed line: the restored early-stopping epoch. Curves end where patience expired.

Table 4 — Convergence summary. The gradient norm falls to roughly half its peak in every run; none stopped at the 300-epoch ceiling.

Split	Epochs run	Best epoch	Peak grad. norm	Final grad. norm	Best val. MSE	Time (s)
0	53	28	1.61	0.78	1.07	5.3
1	69	44	1.56	0.69	1.04	6.1
2	84	59	1.52	0.61	1.16	7.3
3	74	49	1.58	0.64	1.07	7.0
4	58	33	1.74	0.75	1.01	5.3

The five runs are closely matched: 53–84 epochs, best weights at epoch 28–59, and best validation MSE within 1.01–1.16. The gradient norm plateaus rather than reaching zero, which is expected: dropout and mini-batch sampling keep a stochastic floor under it, so the diagnostic of convergence is the decay and flattening, not a vanishing value.

7. Generalization error

Each model is scored once on its own test set, after early stopping. Figure 2 and Table 5 report the mean over the five splits with the standard deviation as the error bar; individual splits are plotted as dots. The reference is a predictor that always returns the training mean, which isolates how much the sequences actually contribute.

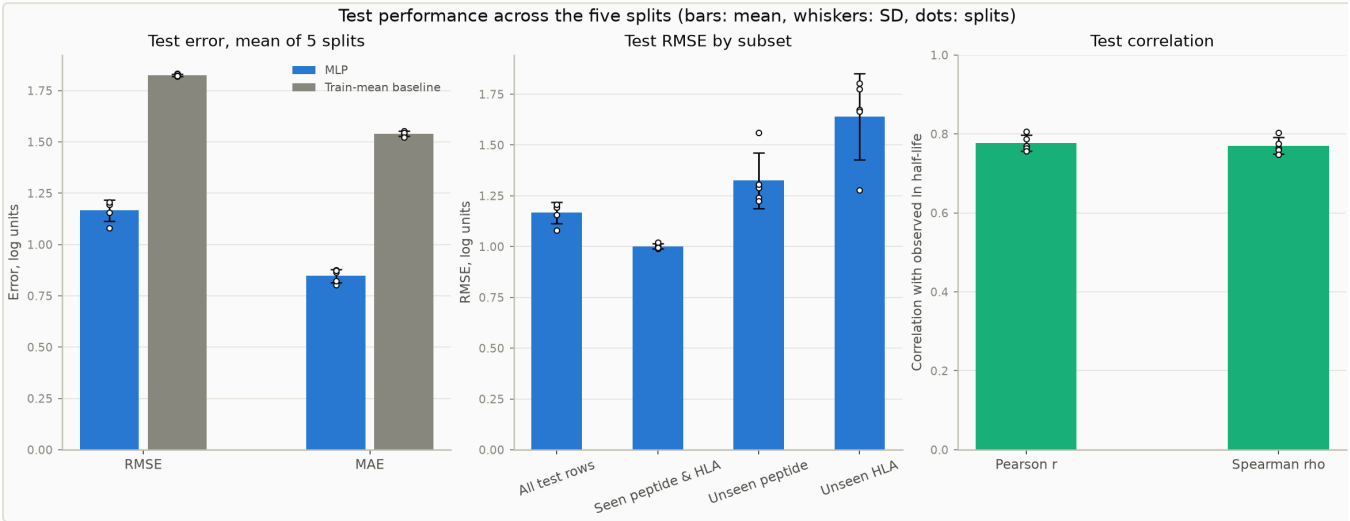


Figure 2. Test performance across the five splits. Bars are means, whiskers standard deviations, dots the individual splits. Left: error against the train-mean baseline. Middle: RMSE by test subset. Right: rank and linear correlation.

Table 5 — Test metrics in natural-log units of hours, mean $\pm$ SD over the five splits ($n = 5$).

Metric	Split 0	Split 1	Split 2	Split 3	Split 4	Mean $\pm$ SD
RMSE	1.080	1.194	1.194	1.207	1.157	1.166 $\pm$ 0.052
MAE	0.802	0.863	0.875	0.873	0.822	0.847 $\pm$ 0.033
Bias	0.049	-0.208	-0.010	0.008	-0.079	-0.048 $\pm$ 0.101
Pearson r	0.807	0.769	0.764	0.756	0.787	0.777 $\pm$ 0.020
Spearman ρ	0.803	0.764	0.759	0.748	0.776	0.770 $\pm$ 0.021
Baseline RMSE	1.820	1.832	1.830	1.822	1.821	1.825 $\pm$ 0.006

The model cuts RMSE by 36% relative to the baseline and ranks complexes with $\rho = 0.77$, so the stacked sequences carry substantial signal. Mean bias is near zero and varies little by split (-0.21 to $+0.05$), so there is no systematic offset; the calibration problem is conditional rather than global, as section 8 shows.

Breakdown by test subset

Test rows are grouped by whether their peptide and HLA pseudosequence were present anywhere in the training set. This is the comparison the constrained splits were designed to support.

Table 6 — Test RMSE by subset, mean $\pm$ SD over splits. Subset sizes vary by split because holdout counts are minimums, not targets.

Subset	Rows per split	RMSE, mean $\pm$ SD	vs. seen
Seen peptide and HLA	5,695-6,389	1.001 $\pm$ 0.012	—
Unseen peptide	963-1,271	1.325 $\pm$ 0.137	+32%
Unseen HLA	1,126-1,837	1.640 $\pm$ 0.211	+64%
All test rows	8,450	1.166 $\pm$ 0.052	+16%

Two results stand out. Generalization to **unseen peptides** costs 32% RMSE: with thousands of distinct 9-mers in training, the network learns position-wise peptide chemistry that transfers, if imperfectly. Generalization to **unseen HLAs** costs 64% and reaches 1.80 in the worst split, approaching the 1.83 of a model that ignores the sequences entirely. Performance on seen combinations is by contrast remarkably stable, RMSE 1.001 $\pm$ 0.012 across splits, so nearly all the split-to-split variation in the headline number comes from which alleles were withheld.

The unseen-HLA figure should be read as a weak estimate, not a stable benchmark: 5 held-out alleles per split, with overlapping test sets between splits, means the SD of 0.59 understates the true uncertainty across possible allele choices.

8. Structure of the error

Figure 3 plots every test prediction against its measurement, pooled over the five splits. The vertical stripes at the left are the discreteness of the assay: half-lives are reported to one decimal, so the measured axis has few distinct values there, the leftmost being the zero rows at $\ln(0.1)$.

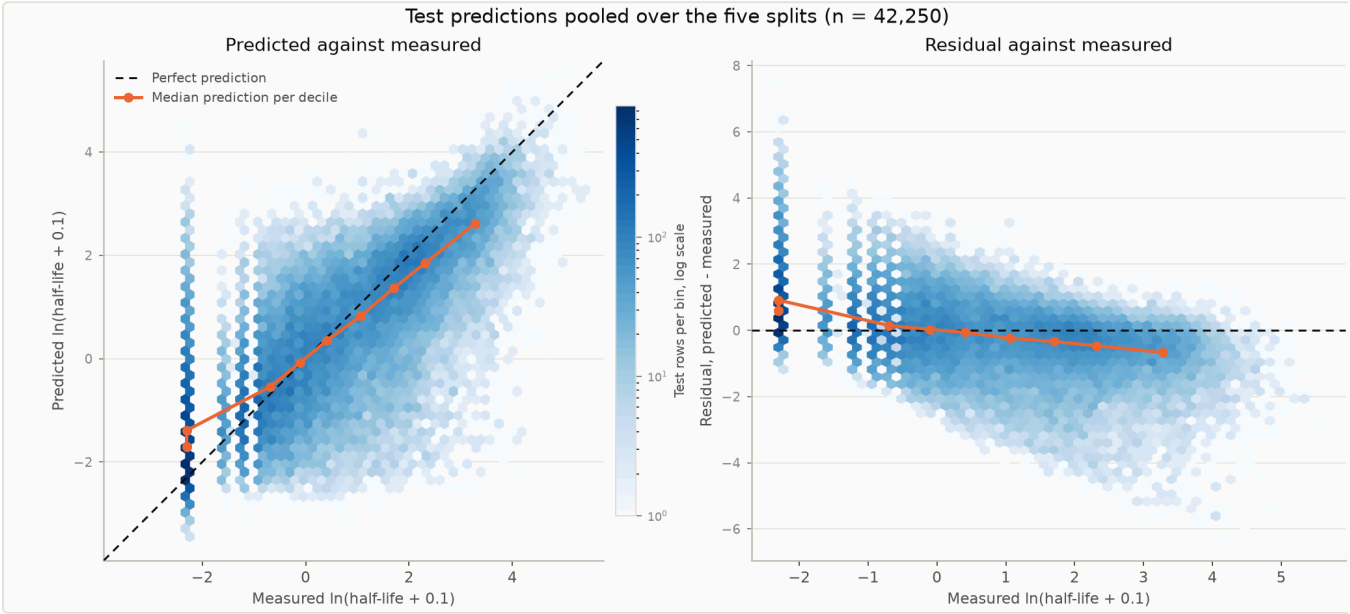


Figure 3. Left: predicted against measured log half-life, 42,250 test rows binned hexagonally with a logarithmic count scale; dashed line is equality, ■ the median prediction per decile of the measurement. Right: the same data as residuals. The median line crossing from above the dashed line to below it is the shrinkage described below.

The cloud follows the diagonal but is visibly flatter than it, which is the signature of regression to the mean under a squared-error loss: the model compresses its predictions toward the centre of the distribution. The median prediction per decile sits above equality at the low end and below it at the high end, crossing near the middle of the range.

Table 7 — Squared error by measured half-life, mean $\pm$ SD over splits. Shares sum to 100% within each split.

Measured half-life	% of test rows	% of squared error	RMSE	Mean signed error
= 0 h	20.1	24.7 ± 6.5	1.282	+0.847
(0, 0.5] h	17.9	12.4 ± 1.6	0.969	+0.300
(0.5, 2] h	22.5	17.5 ± 2.2	1.026	-0.036
(2, 8] h	21.0	20.9 ± 1.7	1.166	-0.510
> 8 h	18.5	24.5 ± 6.9	1.336	-0.846

Error is distributed far more evenly across the range than under the earlier $\varepsilon = 0.001$, where the zero rows alone produced 47% of the squared error. What remains is a U shape: the two extremes of the distribution are hardest, and the signed errors confirm the shrinkage — zero-stability complexes are predicted too stable by +0.85 log units, complexes above 8 h too unstable by -0.85, with the crossover in the (0.5, 2] stratum.

Read as a stability call

Thresholding the regression turns signed error into false positives (predicted stable, measured unstable) and false negatives. Which error dominates depends entirely on where the threshold is placed.

Table 8 — Stability call at three thresholds, mean counts and rates over the five splits of 8,450 test rows.

Threshold	False positives	False negatives	FP rate	FN rate	Precision	Recall
> 0.5 h	837	804	26.1%	15.4%	0.842	0.846
> 1 h	848	823	20.2%	19.4%	0.802	0.806
> 2 h	707	801	13.8%	24.1%	0.783	0.759

At a 1-hour threshold the two error types are balanced. Below it false positives dominate, as the over-predicted zero and near-zero rows cross the line; above it false negatives dominate, as shrinkage pulls genuinely stable complexes below it. Screening for strongly stable complexes (> 2 h) therefore loses about a quarter of them, which is the practical cost of the shrinkage and the main target for future work.

9. Choosing the log offset ϵ

ϵ is not a free parameter: it decides how far the 20% of zero-stability rows sit below the measurable ones, and therefore how much of the loss they absorb. Four values were trained on all five splits with every other setting fixed.

Comparison is not straightforward, because ϵ changes the units of the target and so rescales RMSE directly — raw RMSE falls from 2.42 to 0.64 across this sweep almost entirely by rescaling. Two transform-independent measures are used instead: Spearman correlation, invariant under any monotone transform, and an RMSE computed after mapping every model's predictions back to hours and into one fixed space, $\ln(t + 0.1)$.

Table 9 — ϵ sweep, mean $\pm$ SD over the five splits. The last column is the median predicted half-life for complexes measured at exactly zero.

ϵ	Spearman ρ	RMSE, common space	FN rate > 2 h	FP rate > 2 h	Median prediction at $t = 0$
0.001	0.742 $\pm$ 0.015	1.292 $\pm$ 0.046	27.5%	14.7%	0.007 h
0.01	0.756 $\pm$ 0.025	1.224 $\pm$ 0.064	28.0%	12.6%	0.025 h
0.1 (adopted)	0.770 $\pm$ 0.021	1.164 $\pm$ 0.052	24.1%	13.9%	0.083 h
1.0	0.769 $\pm$ 0.020	1.155 $\pm$ 0.050	23.5%	14.0%	0.187 h

The splits are paired, so per-split differences carry more information than the standard deviations. Against $\epsilon = 0.001$, both 0.1 and 1.0 improve Spearman on *all five* splits (+0.028 on average), improve common-space RMSE on all five (−0.13), and reduce the > 2 h false-negative rate on all five (−3.4 and −4.0 points). The mechanism is the one Table 7 makes explicit: with a larger ϵ the zero rows stop monopolising the loss and the network spends its capacity on the measurable range.

Between 0.1 and 1.0 there is nothing to choose statistically — the differences are well inside the split-to-split spread. **0.1 was adopted** on two grounds beyond the metrics:

- **It matches the assay.** Half-lives are reported to one decimal, so 0.1 h is the resolution and plausibly the detection limit. $\epsilon = 0.1$ states that zero and 0.05 h are indistinguishable, which is true of the data.
- **$\epsilon = 1$ flattens the low range.** Since $\ln(1 + t) \approx t$ for small t , the target becomes effectively linear in hours for the 58% of rows below 1 h, discarding the log resolution that separates 0.1 from 0.5 h. Its median prediction for a truly-zero complex is 0.19 h, nearly twice the assay resolution.

Caveat: the architecture and learning rate were tuned for none of these settings, so some of the gap might close under retuning. More fundamentally, tuning ϵ is a workaround for a censoring problem — if zeros mean "below detection", the principled treatment is a censored likelihood that reads them as $t < 0.1$ rather than $t = 0.1$.

10. Limitations and next steps

1. **Censoring is handled by a constant, not a model.** $\epsilon = 0.1$ is an informed choice (section 9) but still asserts a value for measurements that have none. A censored likelihood, or the bounded transform s

= $2^{-t_0/t}$ used by NetMHCstabpan, would treat the detection limit explicitly and make results comparable with the literature.

2. **Predictions are shrunk toward the mean.** The ± 0.85 log-unit bias at the extremes (Table 7, Figure 3) costs about a quarter of strongly stable complexes at a 2-hour threshold. If a specific operating threshold matters, a classifier at that threshold, or a two-stage binder-then-stability model, would likely beat thresholding this regressor.
3. **Validation is optimistic.** Holding out whole peptides and HLAs for validation, mirroring how the test sets are built, would stop training at a point tuned for unseen sequences; this most plausibly affects the unseen-HLA result.
4. **One-hot ignores residue similarity.** BLOSUM62 encoding is a drop-in replacement at the same input width and is the most likely single improvement for unseen HLAs.
5. **No hyperparameter search.** Widths, dropout, learning rate and weight decay are untuned defaults. Any search must use the validation sets only.
6. **Few HLAs.** With 74 pseudosequences, allele-level generalization cannot be estimated precisely from 5 held-out alleles per split; leave-one-allele-out over all 74 would measure it properly, at 74 training runs.
7. **Date-recovered targets.** 11,907 of 28,166 half-lives (42%) were stored as dates in the source workbook and recovered as `day.month` by the splitting script. The recovery is consistent with one-decimal values, but it is lossy in principle and worth confirming against the original publication.

11. Reproducing this work

```
cd peptide-HLA-stability
.venv/bin/python split_dataset.py DATA/rasmussen_et_al_dataset.xlsx --thresholds 0.05 0.1
.venv/bin/python train_mlp.py --splits-dir DATA --output-dir RESULTS
.venv/bin/python analyze_errors.py --splits-dir DATA --results-dir RESULTS
```

Figures 1 and 2 and `metrics.json` come from the second command, Figure 3 and the error decomposition from the third. $\epsilon = 0.1$ is the default; the sweep in section 9 is reproduced by passing `--epsilon` and a separate output directory.

Table 10 — Files produced.

Path	Contents
DATA/training_i.npz, testing_i.npz	The five splits, $i = 0 \dots 4$, one array per column
DATA/split/	Splitting script's own workbooks, metadata and log-distribution plots
RESULTS/mlp_split_i.pt	Restored best weights plus the target transform constants
RESULTS/metrics.json	Full configuration, per-epoch histories and every metric in this report
RESULTS/convergence.png, test_error.png	Figures 1 and 2
RESULTS/predicted_vs_true.png	Figure 3
RESULTS/error_decomposition.png, .json	Strata and stability calls behind Tables 7 and 8

Environment: Python 3.14.2, PyTorch 2.14.1, NumPy 2.5.3, pandas 3.0.6, Matplotlib 3.11.2, CPU only.
Seeds are fixed (split i uses seed i for the validation draw, weight initialization and batch order), so the numbers above reproduce exactly.

Generated from RESULTS/metrics.json; all metrics quoted are taken from that file. Source: RESULTS/report.html.